UTHM INDUSTRIAL VISIT

Summary of the Visit

Recently, our team conducted an industrial visit to Universiti Tun Hussein Onn Malaysia (UTHM) as part of our enterprise system project. The visit aimed to gain exposure to UTHM's digital environment and to collect relevant information on their enterprise-level systems and digital transformation initiatives.

This programme included a visit to the UTHM Digital Centre and a briefing on the university's tehcnology infrastructure and digital transformation journey. The session provided insights into how digital solutions are implemented to support academic, administrative and organizational processes.

Reflections from the visit

This visit enhanced my understanding of how enterprise systems and digital infrastructure support university operations at an organizational level. Learning about UTHM's digital transformation journey highlighted the importance of strategic planning and alignment between technology and institutional goals. The experience made me realize how important it is to undergo every crucial steps carefully in designing an enterprise architecture.

MUHAMMAD AFIQ DANISH (A23CS0118)

This visit revealed how data center infrastructure impacts daily university operations. Seeing UTHM's digitalization journey taught me that digital transformation takes careful planning and time. This experience motivated me to pursue deeper knowledge in IT infrastructure management and appreciate how thoughtful system design directly improves user experiences.

NAJMA SHAKIRAH (A23CS0140)

The industrial visit to UTHM bridged the gap between academic theory and real-world application. In addition to the security protocols needed to protect sensitive data, we gained a deeper understanding of the physical realities of operating a data center.This experience made me realize how critical enterprise architecture is to my career and how to prepare for it through careful planning, resilience, and robust infrastructure.

LING YU QIAN (A23CS0301)

This visit made me realized the extent to which UTHM enterprise systems are backed by technologies and infrastructure in an actual industry environment. I was introduced to their system access management and architecture planning thereby, increasing my understanding of the necessity of a structured design and security, This visit has also been a great opportunity making me more familiar with the connections between academic theories and real-world practices.

NABIL AFLAH BOO (A23CS0252)

Technologies at UTHM

While visiting UTHM, we were given a chance to go around their data center and see the technology infrastructure they have installed. The data center has been built up with hot and cold containment to enhance cooling effectiveness and thereby reduce power consumption in total. It also helps conserve the proper heat for hard IT tools and makes system's operation much less prone to errors. As part of its technology stack, UTHM has been employing a three-tier model comprising presentation, application, and database layers. This architectural design gives better scalability, security, and performance to their corporate systems.

The facility features a three-tier electrical architecture with dual independent power feeds. In case of a power failure, an Uninterruptible Power Supply (UPS) ensures continuous operation until the generator set activates, minimizing downtime. The Monitoring Room serves as the data center's control hub for monitoring environmental and equipment status. The Staging Room is designated for pre-installation equipment preparation and validation by vendors. Additionally, a multi-layered cybersecurity strategy is employed, utilizing two firewalls and Web Application Security (WAF/WAVS) to protect against cyber threats, thereby enhancing security and ensuring service availability.

Issues Discussed

Physical hardware installation was sometimes problematic due to some servers not fitting into racks according to their specific dimensions. Furthermore, the IT team must constantly manage energy and water consumption in the data center in order to keep it cool and running. Most complaints about user experience revolve around campus WiFi connectivity issues, Single Sign-On (SSO) errors, and password forgetting.

Due to a lack of experience in Enterprise Architecture (EA), the team initially struggled from a strategic perspective. As well as complicated logical hurdles, there were also LAN wiring issues, IP address configuration issues, and IPv6 integration issues. Additionally, maintaining the continuous operation of the servers requires a significant amount of financial investment and manpower.

Another distinct challenge is managing system loads, especially during peak times like course registration when there are around 1,000 simultaneous users. Last but not least, the department is currently facing the technical challenge of migrating from a legacy client-server platform to a fully web-based system, a lengthy and complex process requiring meticulous phased planning.